Mahdiyeh Valian

Data Scientist

CONTACT

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Iran-Tehran

Linkdin

Portfolio

skills

- Python
- C/C++
- C#
- Data Analysis
- VHDL, Verilog

Languages

- Persian: Native
- English: Fluent (IELTS expected in December 2026)

About me

Entry-level data scientist with a strong commitment to learning and developing skills in machine learning and deep learning.
Graduated from Alzahra University, Department of Computer Engineering in Master's Degree, member of the Data mining Lab, with research background in time series and motion sensors signal processing.

EDUCATION

M.Sc. in Computer Engineering, Major in Artificial Intelligence Sep 2019 - Mar 2022
Alzahra University, Tehran, Iran
GPA(Total): 3.19/4.0
Advisor: Dr. Mohammad Reza Keyvanpour

- Research Area: "signal and image processing, human activity recognition, neural networks and machine learning".
- Thesis: "Wearable sensors-based human activity recognition using deep learning" (Thesis GPA: 4.0/4.0)

B.Sc. in Computer Engineering, Major in Hardware Engineering Sep 2012 - Mar 2017
Babol Noushiravni University of Technology, Mazandaran, Iran
GPA(Last two years): 3.0/4.0

Research Interests

- Machine Learning and Deep Learning.
- Time-Series Analysis and Modeling.
- Signal Processing and Pattern Recognition.
- Human Activity Recognition and Wearable Sensing.
- Neural Networks and Representation Learning.

Academic Projects

Shahid Beheshti University Apr 2023 - Jan 2025
Supervisor: Dr. Ali Jahanian

- Remote Side-Channel Analysis Attacks on FPGA-based Multi-tenant Servers using Deep Neural Networks.

MSc Thesis: Alzahra University Feb 2020 - Mar 2023
Supervisor: Dr. MohammadReza keyvanpour

- Title: Wearable sensors-based human activity recognition using deep learning.
- Description: Analyzing time series signals which are obtained from accelerometer and gyroscope sensors for categorizing different activities.

BSc projects: Babol Noshirvani University Jun 2016 - Sep 2017
Supervisor: Dr. Ali Gholami Rudi

- Title: Calculation of Additive Spanners for graph.
- Description: Doing project with programming software of C++;Kinds of graphs:Directed graph, Isomorphic graph, Counting graphs, Complete graph, Complete graph, K-regular graph,Bipartite graph, Complete bipartite graph, digraph, Adjacency Lists, Adjacency matrix.Algorithm for graph process : Warshall's algorithm,The lightest path, Dijkstra's algorithm, Floyd's algorithm, Thelightest spanning tree, Kruskal's algorithm, Prim's algorithm, the lightest Hamiltonian circuit, the annealing algorithm,Karp-held heuristics, the Hungarian algorithm, The Ford-Fulkerson, Breadth-first search, Depth-first search, Hadoop Mapreduce, Classical routing, Spanner,Additive Spanner.

Work Experience

Speed Energy Aryanic — Mar2021 - Jan 2022

C/C++ Programming at Microcontroller Programming Department (Internship)

- UART Serial Interface Programming

Development of a chained UART interfaces among multiple Arduino mini pro development boards.

Telecommunication Company — Jun 2026 - Aug 2026

Network Management (Internship)

- experience in switch room, operating maintenance room, Main Distribution Unit, Digital Distribution Unit.

Course Projects

- improving DBScan algorithm by genetic algorithm using tournoment selection method implementing on Iris dataset (machine learning MSc Course)
- Calculating the similarity between words and sentences using WordNet database (natural language processing MSc Course)
- sleep stage scoring by implementing sleepEEGNet a sequence to sequence deep learning method (speech and speaker recognition MSc Course)
- implementing Genetic-shuffled Frog-Leaping Algorithm for Text Document Clustering using the “20Newsgroup” dataset (evolutionary computing MSc Course)

References

- Dr. Mohammad Reza Keyvanpour
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  Scholar page
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